ALO–RATS: Accelerated, Long-Horizon Planning and Online Adjustment for Robot-Assisted Tissue Surgery

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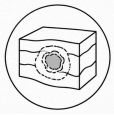
* Equal contribution † Equal advising

Brain Tool Laboratory

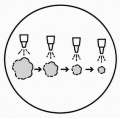
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Contributed abstract

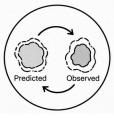
Motivations



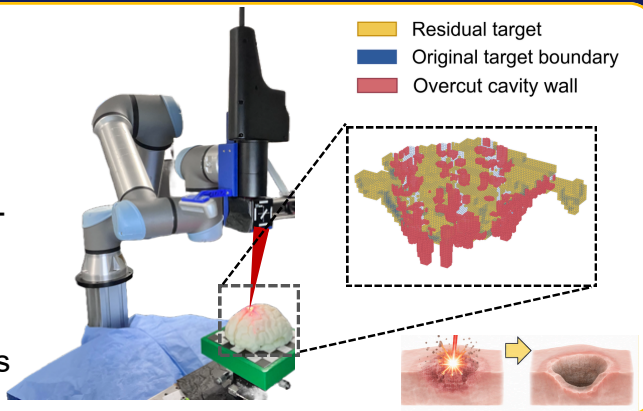
➢ **Volumetric Resection:** Precise 3D target removal while preserving critical tissue.



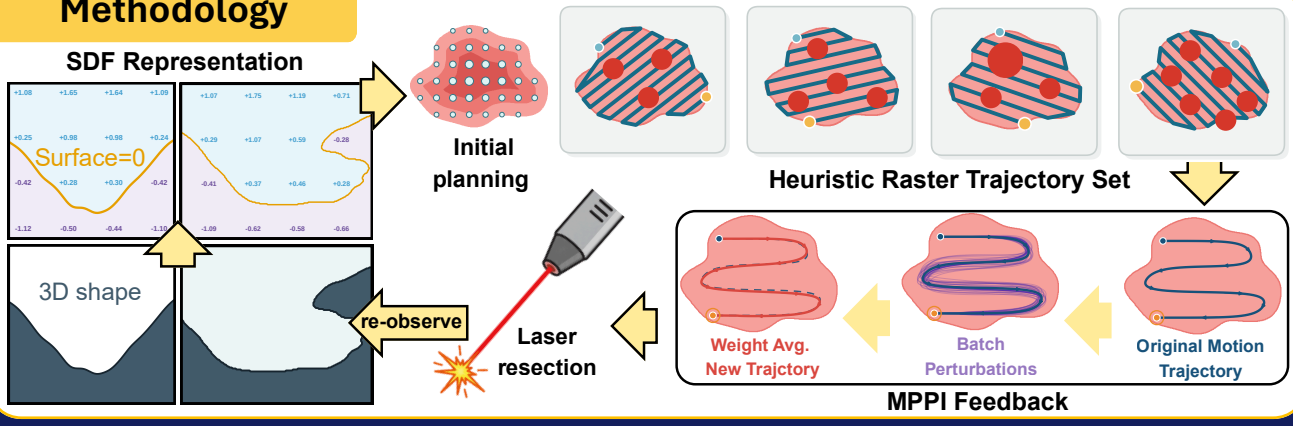
➢ **Long-Horizon Planning:** Irreversible laser actions make late-stage residual tissue difficult to resolve.



➢ **Online Plan Repair:** Feedback corrects model and execution errors during resection.



Methodology



Experiment Result

8.6× Faster Planning

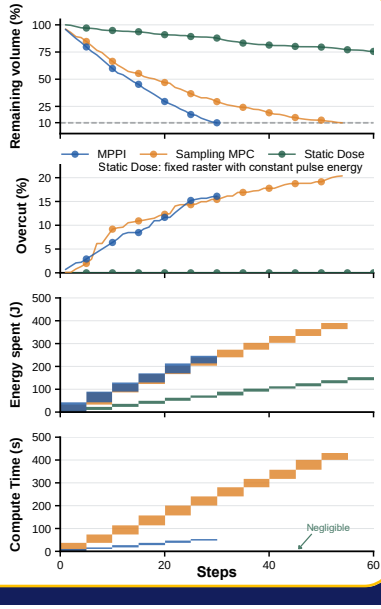
52.3% Fewer Pulses

38.3% Less Energy

Tumor Test



Metric–Case	1	2	3	4	5
Target (mm^3)	420	215	245	388	1284
Remaining (%)	2.98	2.67	2.96	2.95	2.99
Overcut (%)	18.57	21.01	21.87	24.76	14.09
Healthy (mm^3)	77.98	45.17	53.58	96.08	180.89
Pulses	236	148	295	319	844
Time (min)	12.00	6.09	21.09	22.59	68.63



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